

Convex and Quasiconvex Functions

Let $f : D \rightarrow \mathbb{R}$ for some nonempty convex subset D of $\mathbb{R}^n$. The function f is *convex* if, for any x and y in D and $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$. f is *quasiconvex* if, for any x, y in D and $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$.

The following facts are immediate: (a) If f is convex, then it is quasiconvex. (b) The sum of convex functions is convex. (c) If f is quasiconvex and $T : \text{Range}(f) \rightarrow \mathbb{R}$ is strictly increasing, then $g = T \circ f$ is quasiconvex. (Proofs?)

If f is a C^2 function on an open set D (whether quasiconvex or not), its *Hessian* at $x \in D$, $D^2f(x)$, is a symmetric $n \times n$ matrix whose i, j element is $f_{ij}(x)$.

Fact. Suppose that f is C^2 on an open convex set D . The following are equivalent.¹

- f is convex.
- For every $x \in D$, $D^2f(x)$ is (symmetric and) positive semidefinite: that is, for any $z \in \mathbb{R}^n$, it follows that $z^T \times D^2f(x) \times z = \sum_i \sum_j f_{ij}(x) z_i z_j \geq 0$.²
- For every $x \in D$, the principal minors of $D^2f(x)$ are all nonnegative.³

For the case of $n = 2$, the last criterion asserts that, for every $x \in D$, $f_{11}(x) \geq 0$, $f_{22}(x) \geq 0$, and $f_{11}(x)f_{22}(x) - f_{12}(x)^2 \geq 0$.

¹For a reference, see Sections 7.2.3 and 7.6 of Sundaram's *A First Course in Optimization Theory*.

²In particular, note that $f_{ii}(x) \geq 0$ for each i .

³A principal minor is the determinant which remains when some rows and the same-numbered columns are eliminated